

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built OCR (including Tesseract) into a production RAG ingestion pipeline in Python, extracting text from scanned and image-based documents to feed the retrieval and grounding layer for LLM-based information extraction.
- Designed a multi-agent system (OpenAI Agents SDK, MCP tool-calling) with vector search and embedding-based retrieval, plus GraphRAG improving LLM response grounding over standard vector RAG for complex, multi-hop queries.
- Deployed and operated production AI workloads on GCP and AWS, hardening the system with a deployment-gated evaluation pipeline enforcing automated data quality and drift checks, blocking 15% of releases that failed validation.

QureAI (Healthcare AI Startup)

Jan 2024 – May 2025

ML Infrastructure Engineer

- Built agentic AI workflows applying LLM-based automation to clinical/medical data in a regulated healthcare product; designed and implemented a Neo4j knowledge graph deployed to production to power graph-aware retrieval.
- Built PySpark data processing jobs for large-scale feature/dataset preparation, and led migration of model serving to async vLLM, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices processing regulated financial transactions for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning, in a domain requiring auditable, traceable transaction records.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL, built to keep agentic systems trustworthy at volume.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed, traceable root cause reports.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, TypeScript, SQL

Document AI & ML Engineering: OCR (Tesseract), Document Extraction, RAG, GraphRAG, Knowledge Graphs (Neo4j), Agentic AI, LangGraph, OpenAI Agents SDK, MCP, Vector Databases (Pinecone, FAISS), PySpark

Backend & APIs: FastAPI, REST APIs, Node.js, PostgreSQL

Cloud: GCP (Cloud Run), AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), Azure, Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer